



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 19/02/2018*

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = S[n] + W[n] = A \cos(2\pi F_0 n - \theta) + W[n],$$

where A and θ are mutually independent random parameters and $F_0 \in (0, 1/2)$ is a-priori known, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_w(e^{j2\pi f}) = \sigma_w^2$, where σ_w^2 is a-priori known. A is a Rayleigh distributed random variable with power $E\{A^2\} = \sigma_A^2$ and θ is uniformly distributed in $[-\pi, \pi)$.

(1) Derive the Signal to Noise power Ratio (SNR) $\gamma \triangleq \frac{E\{S^2[n]\}}{E\{W^2[n]\}}$, the mean value and the autocorrelation function (ACF) of the process $X[n]$.

(2) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$.

(3) Derive the optimal linear estimator of the signal of interest $S[n]$ given the observation of two samples of the process $X[n]$ at past times, i.e. $\hat{S}[n] = h[1]X[n-1] + h[2]X[n-2]$, and express it as a function of the SNR γ . Calculate numerically $h[1]$ and $h[2]$ under the assumption that $\gamma = 1$ and $F_0 = 1/8$.

(4) Derive the Mean Square Error $\sigma_\varepsilon^2 \triangleq E\{(S[n] - \hat{S}[n])^2\}$ of the optimal linear estimator of $S[n]$ as a function of the SNR γ . Calculate numerically $h[1]$ and $h[2]$ under the assumption that $\sigma_A^2 = 2$, $\gamma = 1$ and $F_0 = 1/8$.

Hint: $f_A(a) = \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a)$, $E\{A^2\} = \sigma_A^2$.



Solution

(1) The SNR, the mean value and the ACF are:

$$\gamma \triangleq \frac{E\{S^2[n]\}}{E\{W^2[n]\}} = \frac{\sigma_A^2}{2\sigma_W^2}$$

$$\eta_X[n] = E\{X[n]\} = 0$$

$$R_X[m] = E\{X[n]X[n+m]\} = R_S[m] + R_W[m] = \frac{\sigma_A^2}{2} \cos[2\pi F_0 m] + \sigma_W^2 \delta[m]$$

(2) The pdf can be derived by observing that $X[n]$ can be expressed as follows:

$$\begin{aligned} X[n] &= A \cos(2\pi F_0 n - \theta) + W[n] = A \cos(\theta) \cos(2\pi F_0 n) + A \sin(\theta) \sin(2\pi F_0 n) + W[n] \\ &= X_1 \cos(2\pi F_0 n) + X_2 \sin(2\pi F_0 n) + W[n] \end{aligned}$$

It can be proved that X_1 and X_2 are Gaussian distributed random variables, so also $X[n]$ is a Gaussian random variable, and as consequence, $\mathbf{X}$ is a Gaussian random vector: $\mathbf{X} \in N(\boldsymbol{\eta}_X, \mathbf{C}_X)$, where:

$$\boldsymbol{\eta}_X = E\{\mathbf{X}\} = \mathbf{0}, \quad \mathbf{C}_X = \mathbf{C}_S + \mathbf{C}_W = \mathbf{C}_S + \sigma_W^2 \mathbf{I}, \quad \text{where } [\mathbf{C}_S]_{i,j} = R_S[j-i] = \frac{\sigma_A^2}{2} \cos[2\pi F_0(j-i)].$$

(3) To find optimal linear predictor in AWGN, we have to apply the Orthogonality Principle: the estimation error $\varepsilon[n] = S[n] - \hat{S}[n]$ should be orthogonal to the data $X[n-1]$ and also $X[n-2]$.

$$\min_{\{h[n]\}} \sigma_\varepsilon^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad m=1,2$$

$$\begin{aligned} E\{\varepsilon[n]X[n-m]\} &= E\{(S[n] - \hat{S}[n])X[n-m]\} = E\left\{\left(S[n] - \sum_{k=1}^2 h[k]X[n-k]\right)X[n-m]\right\} \\ &= E\{S[n]X[n-m]\} - \sum_{k=1}^2 h[k]E\{X[n-k]X[n-m]\} \\ &= R_{SX}[-m] - \sum_{k=1}^2 h[k]R_X[m-k] = 0, \quad m=1,2 \end{aligned}$$

Taking into account that in our case $R_{SX}[-m] = R_S[m]$, we finally came up with the linear system of equations:



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$\sum_{k=1}^2 h[k] R_x[m-k] = R_s[m], \quad m=1,2$$

that can be expressed in matrix format:

$$\begin{bmatrix} R_x[0] & R_x[1] \\ R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

which in our case becomes:

$$\begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \frac{\sigma_A^2}{2} \cos[2\pi F_0] \\ \frac{\sigma_A^2}{2} \cos[2\pi F_0] & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} \frac{\sigma_A^2}{2} \cos[2\pi F_0] \\ \frac{\sigma_A^2}{2} \cos[4\pi F_0] \end{bmatrix}$$

$$\frac{\sigma_A^2}{2} \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{\sigma_A^2}{2} \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix}^{-1} \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} \begin{bmatrix} 1+1/\gamma & -\cos[2\pi F_0] \\ -\cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$

$$h[1] = \frac{(1+1/\gamma) \cos[2\pi F_0] - \cos[2\pi F_0] \cos[4\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]}$$

$$h[2] = \frac{(1+1/\gamma) \cos[4\pi F_0] - \cos^2[2\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]}$$

Assumption: $\gamma=1$, $F_0=1/8 \Rightarrow h[1]=0.4041$, $h[2]=-0.1429$.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

(4) The MSE is obtained by applying again the the Orthogonality Principle:

$$\begin{aligned}\sigma_{\varepsilon}^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=1}^2 h[k]E\{S[n]X[n-k]\} \\ &= R_s[0] - \sum_{k=1}^2 h[k]R_s[k] \\ &= \frac{\sigma_A^2}{2} (1 - h[1]\cos[2\pi F_0] - h[2]\cos[4\pi F_0])\end{aligned}$$

Assumption: $\sigma_A^2 = 2$, $\gamma = 1$, $F_0 = 1/4 \Rightarrow \sigma_{\varepsilon}^2 = 0.7143$